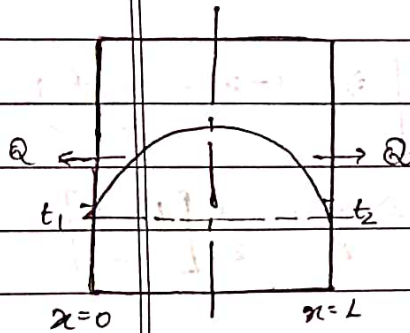


$$t = -\frac{q_g}{2k} x^2 + \frac{q_g L}{2k} x + t_w$$

Heat conduction with
Internal heat generation

$$t = -\frac{q_g}{2k} (L-x)x + t_w$$

Plane wall



To find maximum temp location

$$\frac{dt}{dx} = 0$$

$$\frac{dt}{dx} - \frac{q_g}{2k} (L-2x) = 0$$

$$L-2x=0 \Rightarrow L=2x \text{ or } x=L/2$$

$$t_m = \frac{q_g}{2k} (L-L/2) L/2 + t_w$$

$$t_m = \frac{q_g L^2}{8k} + t_w$$

- Unidirection

- Internal H.G.

$$\nabla^2 t + \frac{q_g}{k} - \frac{L}{\alpha} \frac{\partial t}{\partial z}$$

$$\frac{d^2 t}{dx^2} + \frac{q_g}{k} = 0 \Rightarrow \frac{d^2 t}{dx^2} = -\frac{q_g}{k}$$

$$\frac{dt}{dx} = -\frac{q_g}{k} x + C_1$$

$$t = -\frac{q_g}{2k} x^2 + C_1 x + C_2$$

$$Q = -kA \left[\frac{dt}{dx} \right]_{x=L} = -kA \left[\frac{q_g}{2k} (L-2x) \right]_{x=L}$$

$$= -kA \frac{q_g}{2k} (-L)$$

$$Q = \frac{q_g AL}{2}$$

for both side

$$Q = q_g AL$$

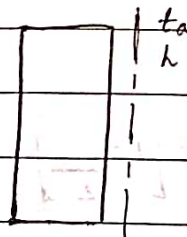
1. Both wall temp same

$$t_1 = t_2 = t_w$$

$$x=0 \quad t=t_w$$

$$x=L \quad t=t_w$$

$$t_w = C_2$$



$$Q = \frac{q_g AL}{2} = hA(t_w - t_a)$$

$$t_w = -\frac{q_g}{2k} L^2 + C_1 L + t_w$$

$$C_1 L = \frac{q_g L^2}{2k} \Rightarrow C_1 = \frac{q_g L}{2k}$$

$$C_1 = \frac{q_g L}{2k}$$

$$\frac{q_g AL}{2hA} = t_w - t_a \Rightarrow t_w = t_a + \frac{q_g L}{2h}$$

$$t = \frac{q_g}{2k} (L-x)x + t_a + \frac{q_g L}{2h}$$

$$t_m = \frac{q_g}{8k} L^2 + t_m + \frac{q_g L}{2h}$$

$$t_m = t_a + q_g \left[\frac{L^2}{8k} + \frac{L}{2h} \right]$$

Both will have diff. temp

$$\frac{dt}{d(x/L)} = \left[\frac{zx+1}{L} \right] [0-1] + [z] \left[\frac{1-x}{L} \right] = 0$$

$$x=0 \quad t = tw_1$$

$$x=L \quad t = tw_2$$

$$x \left[\frac{1-x}{L} \right] - \left[\frac{zx+1}{L} \right] = 0$$

$$tw_1 = C_2$$

$$\Rightarrow x \left[\frac{1-x}{L} \right] = \left[\frac{x+1}{L} \right] \Rightarrow \frac{1-x}{L} = \frac{x+1}{L} \Rightarrow \frac{1-x}{L} = \frac{x+1}{L}$$

$$tw_2 = \frac{-q_g L^2 + C_1 L + tw_1}{2k}$$

$$\frac{2x}{L} = \frac{1-1}{L} = 0 \Rightarrow x = \frac{1}{2} \left[\frac{1-1}{L} \right]$$

$$tw_2 - tw_1 + \frac{q_g}{2k} L^2 = C_1 L$$

$$\frac{x = z-1}{L = 2z}$$

$$\frac{(tw_2 - tw_1)}{L} + \frac{q_g \cdot L}{2k} = C_1$$

$$\frac{t_{max} - tw_2}{tw_1 - tw_2} = \left[\frac{zx(z-1)+1}{2z} \right]$$

$$t = -\frac{q_g}{2k} \frac{x^2}{L} + \frac{(tw_2 - tw_1)}{L} x + \frac{q_g}{2k} Lx + tw_1$$

$$\left[1 - \frac{(z-1)}{2z} \right]$$

$$= \frac{q_g}{2k} (L-x)x + \frac{(tw_2 - tw_1)}{L} x + tw_1$$

$$= \left[\frac{z-1+1}{2} \right] \left[\frac{1-(z-1)}{2z} \right]$$

$$\Rightarrow t - tw_2 = \frac{q_g}{2k} (L-x)x + \frac{(tw_2 - tw_1)}{2} + (tw_1 - tw_2)$$

$$= \frac{z-1+2}{2} \times \frac{2z-z+1}{2z}$$

$$\Rightarrow \frac{t - tw_2}{tw_1 - tw_2} = \frac{q_g}{2k} \times \frac{1}{(tw_1 - tw_2)} \times (L-x)x + 1 - x/L$$

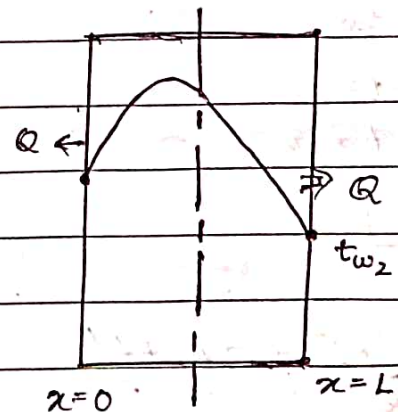
$$\frac{t_{max} - tw_2}{tw_1 - tw_2} = \frac{(z+1)^2}{4z}$$

$$= \frac{q_g}{2k} \times \frac{1}{[tw_1 - tw_2]} \times [Lx - x^2] + \left[\frac{1-x}{L} \right]$$

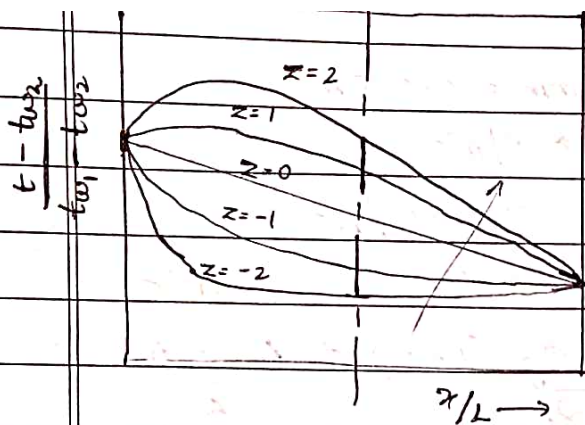
$$= \frac{q_g}{2k} \times \frac{L^2}{tw_1 - tw_2} \times \frac{x}{L} \left[\frac{1-x}{L} \right] + \left[\frac{1-x}{L} \right]$$

$$z = \frac{q_g}{2k} \frac{L^2}{(tw_1 - tw_2)}$$

$$\frac{t - tw_2}{tw_1 - tw_2} = z \cdot \frac{x}{L} \left[\frac{1-x}{L} \right] + \left[\frac{1-x}{L} \right]$$



$$\frac{t - tw_2}{tw_1 - tw_2} = z \cdot \frac{x}{L} \left[\frac{1-x}{L} \right] + \left[\frac{1-x}{L} \right]$$



$$t_{x=1/4} = \frac{q_g}{2k} \left(L - \frac{L}{4} \right) \cdot \frac{L}{4} + t_w$$

$$= \frac{1.2 \times 10^6 \times 3L^2}{32k} + t_w$$

$$= \frac{1.2 \times 10^6 \times 3 \times (0.16)^2}{32 \times 180} + 120$$

$$= 136^\circ\text{C}$$

$$t_{x=3/4} = \frac{q_g}{2k} \left(L - \frac{3L}{4} \right) \cdot \frac{3L}{4} + t_w$$

$$= \frac{q_g}{2k} \cdot \frac{L}{4} \times \frac{3L}{4} + t_w$$

$$= \frac{q_g 3L^2}{32k} + t_w$$

$$= \frac{(1.2 \times 10^6) \times 3 \times (0.16)^2}{32 \times 180} + 120$$

$$= 136^\circ\text{C}$$

The rate of heat generation in a slab of thickness 160 mm ($k = 180 \text{ W/m}^\circ\text{C}$)

is $1.2 \times 10^6 \text{ W/m}^3$. If the temperature of each of the surface of the solid is 120°C . Determine

(i) the temperature at the mid & quarter plane

(ii) the heat flow rate and the temp. gradients at the mid and quarter planes.

$$L = 160 \text{ mm}$$

$$k = 180 \text{ W/m}^\circ\text{C}$$

$$q_g = 1.2 \times 10^6 \text{ W/m}^3$$

$$t_{w1} = t_{w2} = 120^\circ\text{C}$$

$$x = L/2$$

$$x = L/2 \cdot x = 3L/4$$

$$t = \frac{q_g}{2k} (L - x)x + t_w$$

$$\textcircled{1} \quad t_{x=1/2} = \frac{q_g}{2k} (L - 1/2) \cdot \frac{L}{2} + t_w$$

$$= \frac{q_g L^2}{8k} + t_w$$

$$= \frac{(1.2 \times 10^6) \times (0.16)^2}{8 \times 180} + 120$$

$$= 141.33^\circ\text{C}$$

$$Q = q_g \times A \times x \quad (\text{Area} = 1 \text{ m}^2)$$

$$\frac{Q}{A \cdot L} = \frac{Q}{x = L/2} = q_g \cdot \frac{A \cdot L}{2}$$

$$= 1.2 \times 10^6 \times 1 \times 0.16$$

$$= 96000 \text{ W/m}^2$$

$$Q_{x=L/4} = q_g \cdot A \cdot L/4 = 48000 \text{ W/m}^2$$

$$Q = -kA \frac{dt}{dx} \Rightarrow \frac{dt}{dx} = \frac{-Q}{kA}$$

$$\left[\frac{dt}{dx} \right]_{x=L/2} = \frac{-Q_{x=L/2}}{kA}$$

$$= \frac{-96000}{180 \times 1} = -533.33^\circ\text{C/m}$$

$$\left[\frac{dt}{dx} \right]_{x=L/4} = \frac{-Q_{x=L/4}}{kA} = -266^\circ\text{C/m}$$

A metal slab of 25mm thickness having $k = 1600 \text{ W/m}^\circ\text{C}$ is heated with the help of microwave heating. The centre temp. of the slab is maintained at 100°C when surrounding temp is 30°C . The heat transfer coeff. on the surface of slab is $20 \text{ W/m}^2\text{ }^\circ\text{C}$. Find out the microwave heating capacity in W/m^3

$$L = 25 \text{ mm} = 0.025 \text{ m}$$

$$k = 1600 \text{ W/m}^\circ\text{C}$$

$$q_g = 30 \times 10^6 \text{ W/m}^2$$

$$t_{w1} = 180^\circ\text{C}, t_{w2} = 120^\circ\text{C}$$

$$t = \frac{q_g (L-x)}{2h} + \frac{t_{w2} - t_{w1}}{L} x + t_{w1}$$

$$t = \frac{20 \times 10^6 (0.025 - x)}{2 \times 48} + \frac{120 - 180}{0.025} x + 180$$

$$= 5413x - 312500x^2 + 180$$

$$\frac{dt}{dx} = 0 \Rightarrow 5413 - 312500 \times 2x = 0$$

$$x = \frac{5413}{2 \times 312500} = 8.66 \times 10^{-3}$$

$$t_{x=8.66 \times 10^{-3}} = 5413x - 312500x^2 + 180$$

$$= 5413(8.66 \times 10^{-3})$$

$$- 312500(8.66 \times 10^{-3})^2 + 180$$

$$= 203.43^\circ\text{C}$$

$$Q = -KA \left[\frac{dt}{dx} \right]$$

$$Q_{x=0} = -48 \times 1 \times 5413$$

$$= -259800 \text{ N/m}^2$$

$$Q_{x=L} = -KA \left[\frac{dt}{dx} \right]_{x=0.025}$$

$$= -48 \times 1 \times$$

$$[5413 - 2 \times 312500 \times 0.025]$$

$$= 490200 \text{ W/m}^2$$

The temperatures on the two surfaces of a 25mm thick steel plate ($k = 48 \text{ W/m}^\circ\text{C}$) have a uniform volumetric heat generation of $30 \times 10^6 \text{ W/m}^3$ are 180°C and 120°C .

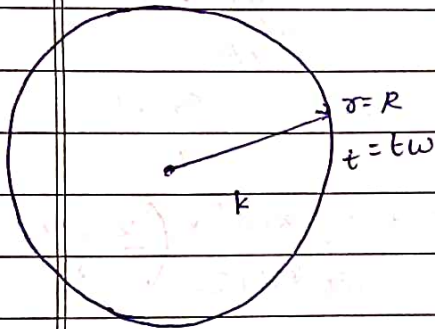
Determine (i) temp. distribution across the plate (ii) the value & position of max. temp. (iii) the flow of heat from each surface of the plate.

$$\alpha = 44 = 48000 \text{ W/m}^2$$

$$Q = -kA \left[\frac{dt}{dx} \right]$$

$$\frac{dt}{dx} = -Q/kA$$

Cylinder with Uniform HG



$$t = \frac{q_g}{k} \cdot \frac{r^2}{4} + C_1 \ln r + C_2$$

$$t = \frac{q_g}{4k} \cdot r^2 + C_1 \ln r + C_2$$

Heat generated = Heat conducted

$$q_g \times (\pi R^2 L) = -k \times 2\pi R \times L \times \left[\frac{dt}{dr} \right]_{r=R}$$

$r=R$

$$\left[\frac{dt}{dr} \right]_{r=R} = \frac{-q_g}{2k} R \quad \text{--- (1)}$$

$$\left[\frac{dt}{dr} \right]_{r=R} = \frac{-q_g}{2k} \times \frac{R}{2} + \frac{C_1}{R} \quad \text{--- (2)}$$

$G=0$

$$\frac{\partial^2 t}{\partial r^2} + \frac{1}{r} \frac{\partial t}{\partial r} + \frac{1}{r^2} \frac{\partial^2 t}{\partial \theta^2} + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial t}{\partial \tau}$$

$$r=R, \quad \tau = t_w$$

→ Unidirection

→ Internal HG

→ Steady

$$t_w = \frac{-q_g}{4k} \cdot R^2 + 0 + C_2$$

$$C_2 = t_w + \frac{q_g R^2}{4k}$$

$$\left(\frac{\partial^2 t}{\partial r^2} + \frac{1}{r} \frac{\partial t}{\partial r} \right) + \frac{q_g}{k} = 0$$

$$\frac{1}{r} \cdot \frac{d}{dr} \left(r \frac{dt}{dr} \right) = -\frac{q_g}{k}$$

$$t = \frac{-q_g}{4k} r^2 + t_w + \frac{q_g R^2}{4k}$$

$$\frac{d}{dr} \left(r \frac{dt}{dr} \right) = -\frac{q_g r}{k}$$

$$t = t_w + \frac{q_g (R^2 - r^2)}{4k}$$

$$t_{\max} = t_w + \frac{q_g R^2}{4k}$$

Integrating both sides w.r.t r

$$r \frac{dt}{dr} = -\frac{q_g}{k} \left[\frac{r^2}{2} \right] + C_1$$

$$\frac{t - t_w}{t_{\max} - t_w} = \frac{R^2 - r^2}{R^2} = 1 - \left(\frac{r}{R} \right)^2$$

Integrating again

$$\frac{dt}{dr} = -\frac{q_g}{k} \times \frac{r}{2} + \frac{C_1}{r}$$

Heat generated = Heat conducted

$$q_g \times \pi R L = h \times 2\pi R L \times (t_w - t_a)$$

$$\Rightarrow \frac{q_g \times R}{2h} = t_w - t_a$$

Integrating again

$$t_w = \frac{q_g R}{2h} + t_a$$

current carrying electrical conductor.

I, R, q, L, A

$$Q = q \times A \times L, \quad R = \frac{\rho \times L}{A}$$

$$A = I^2 R$$

$$Q = I^2 \rho \times \frac{L}{A}$$

$$q = I^2 \rho \times \frac{L}{A}$$

$$q = \left(\frac{I}{A} \right)^2 \times e$$

$$q = \left(\frac{I}{A} \right)^2 \times \rho$$

$$= \left[\frac{300}{\frac{\pi}{4} \times (2.5 \times 10^{-3})^2} \right]^2 \times 70 \times 10^{-8}$$

$$= 2.614 \times 10^9 \text{ W/m}^3$$

$$t_{\max} = t_w + \frac{q \cdot R^2}{4k}$$

$$\text{where } t_w = t_a + \frac{q \cdot R}{2k}$$

$$= 50 + \frac{2.6 \times 10^9 \times (2.5 \times 10^{-3})^2}{2 \times 4000 \times 20}$$

$$= 456.25^\circ \text{C}$$

$$\therefore t_{\max} = 456.25 + \frac{2.6 \times 10^9}{2 \times 4000 \times 20} \times \left[\frac{2.5 \times 10^{-3}}{2} \right]^2$$

$$= 507.03^\circ \text{C}$$

A current of 300 A passes through a stainless steel wire of 2.5 mm diameter and $k = 20 \text{ W/m}^\circ \text{C}$

The resistivity of the wire is $70 \times 10^{-8} \Omega \text{m}$

and the length of the wire is 2 m

If the wire is submerged in a fluid maintained at 50°C and the convective heat transfer coefficient

and wire surface is $4000 \text{ W/m}^2 \text{ } ^\circ \text{C}$,

Calculate the steady state temp. at the center and at the surface of wire.

$$I = 300 \text{ A}$$

$$d = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$$

$$k = 20 \text{ W/m}^\circ \text{C}$$

$$\rho = 70 \times 10^{-8} \Omega \text{m}$$

$$L = 2 \text{ m}$$

$$t_a = 50^\circ \text{C}$$

$$h = 4000 \text{ W/m}^2 \text{ } ^\circ \text{C}$$

$$t_{\max} = ?$$

$$t_w = ?$$

1 m long nichrome wire of

resistivity, $1 \mu \Omega \text{m}$ is to

dissipate power of 10 kW in the surrounding fluid

which is at 80°C . Find the

diameter of the wire if the

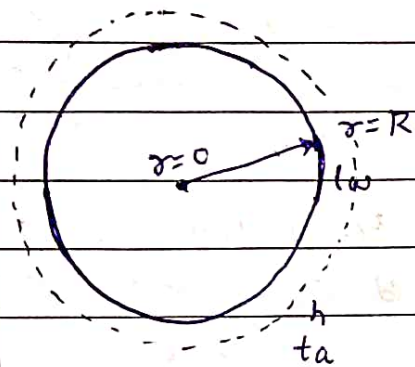
maximum operating temperature

of the wire 1000°C

$$h = 1000 \text{ W/m}^2 \text{ } ^\circ \text{C}$$

$$k = 60 \text{ W/m}^\circ \text{C}$$

Sphere with Uniform
Heat generation



when $r=0 \Rightarrow C_2 = 0$

$$r=R, t=t_w$$

$$Rt_w + \frac{q_g \cdot R^3}{6k} = C_1 R$$

$$C_1 = t_w + \frac{q_g \cdot R^2}{6k}$$

$$t + \frac{q_g r^2}{6k} = C_1$$

$$t + \frac{q_g r^2}{6k} = t_w + \frac{q_g R^2}{6k}$$

$$t = t_w + \frac{q_g R^2}{6k} - \frac{q_g r^2}{6k}$$

$$t = t_w + \frac{q_g (R^2 - r^2)}{6k}$$

→ unidirection

→

$$t_m = t_w + \frac{q_g R^2}{6k}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dt}{dr} \right) + \frac{q_g}{h} = 0$$

$$\frac{1}{r^2} \left(2r \frac{dt}{dr} + r^2 \frac{d^2 t}{dr^2} \right) + \frac{q_g}{k} = 0$$

$$\frac{2}{r} \frac{dt}{dr} + \frac{d^2 t}{dr^2} + \frac{q_g}{h} = 0$$

$$2 \frac{dt}{dr} + r \frac{d^2 t}{dr^2} + \frac{q_g \cdot r}{k} = 0$$

$$r \frac{d^2 t}{dr^2} + \frac{dt}{dr} + \frac{dt}{dr} + \frac{q_g r}{k} = 0$$

$$\frac{d}{dr} \left(r \frac{dt}{dr} \right) + \frac{dt}{dr} + \frac{q_g r}{k} = 0$$

$$r \frac{dt}{dr} + t + \frac{q_g \cdot r^2}{k \cdot 2} = C_1$$

$$\frac{d}{dr} (rt) + \frac{q_g \cdot r^2}{k \cdot 2} = C_1$$

$$rt + \frac{q_g r^3}{k \cdot 6} = C_1 r + C_2$$

$$Q = -kA \left[\frac{dt}{dr} \right]_{r=R}$$

$$= -k(4\pi R^2) \times \frac{d}{dr} \left\{ t_w + \frac{q_g (R^2 - r^2)}{6k} \right\}$$

$$= -k(4\pi R^2) \times \left[\frac{q_g \times -2r}{6k} \right]_{r=R}$$

$$= -k \cdot 4\pi R^2 \times \frac{q_g}{3k} \times -R$$

$$Q = \frac{4\pi R^3 q_g}{3}$$

$$\frac{4\pi R^3 \cdot q_g}{3} = h \times 4\pi R^2 (t_w - t_a)$$

$$\frac{q_g R}{3h} = t_w - t_a$$

$$t_w = t_a + \frac{q_g R}{3h}$$

An approximately spherical shaped orange ($k = 0.23 \text{ W/m}^\circ\text{C}$), 90 mm in diameter undergoes ripening process and generate 5100 W/m^3 of energy. If external surface of the orange is at 8°C , determine

(a) temperature at the center of orange

(b) Heat flow from the outer surface of the orange

$$k = 0.23 \text{ W/m}^\circ\text{C}$$

$$R = 45 \text{ mm} = 45 \times 10^{-3} \text{ m}$$

$$q_g = 5100 \text{ W/m}^3$$

$$t_w = 8^\circ\text{C}$$

$$t_m = t_w + \frac{q_g R^2}{6k}$$

$$8 + \frac{5100 \times (45 \times 10^{-3})^2}{6 \times 0.23}$$

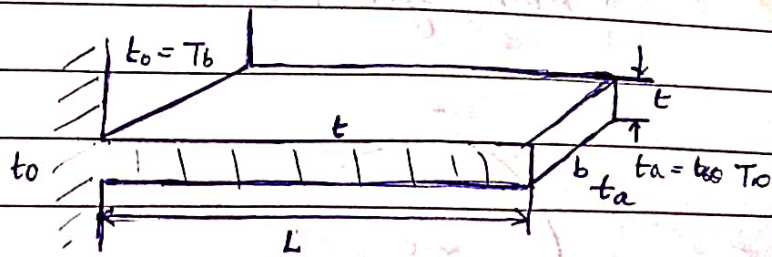
$$Q = \frac{4\pi}{3} = 15.48^\circ\text{C}$$

$$Q = \frac{4\pi R^3 q_g}{3}$$

$$= \frac{4 \times \pi (45 \times 10^{-3})^3 \times 5100}{3} = 1.946 \text{ W}$$

FIND

Pg 61



$$P = \text{perimeter} = 2(b + t)$$

$$A = \text{cross sectional area} = b \times t$$

$$Q = -kA \left[\frac{dt}{dx} \right]$$

$$\frac{d^2 t}{dx^2} - \frac{hP}{kA} (t - t_a) = 0$$

$$\theta = t - t_a \Rightarrow \frac{d^2 \theta}{dx^2} = \frac{d^2 t}{dx^2}$$

$$m = \sqrt{\frac{hP}{kA}}$$

$$\frac{d^2 \theta}{dx^2} - m^2 \theta = 0$$

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

Q1) Calculate the amount of energy required to solder together two very long piece of pair copper wire, 1.5 mm in diameter with solder that melts at 190°C . The wires are positioned vertically in air at 20°C . Assume that the heat transfer coefficient on the wire surface is 20 W/m^2 and thermal conductivity of the wire alloy is $330 \text{ W/m}^\circ\text{C}$.

Q2)

Find out the amount of heat transferred through an iron fin of length 50mm, width 100mm and thickness 5mm, assume $k = 210 kJ/mhr^{\circ}C$ and $h = 42 kJ/m^2hr^{\circ}C$. For the material of the fin and the temperature at the base of fin as $80^{\circ}C$. Also determine the temperature at tip of the fin if atmospheric temperature is $20^{\circ}C$.

Q2) A motor body is 360mm in diameter and 240mm long. Its surface temperature should not exceed $55^{\circ}C$ when dissipating 340W. Longitudinal fins of 15mm thickness and 40mm height are proposed. The convection coeff. is $40 W/m^2^{\circ}C$ ($k = 40$). Atm. temp = $30^{\circ}C$. Find the no. of fins required.